

Lesson #2 : Practical Machine Learning

Human AI Interaction

Motahhare Eslami

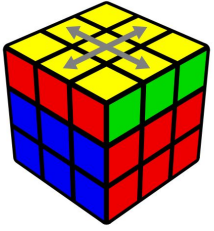
Machine Learning

A “machine” that is able to improve based on past experience *without* explicit human programming on how to improve each time.



"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E ."

-- Tom M. Mitchell



Reminder: AI That Isn't Machine Learning

- Search-Based Approaches
 - A* search, for instance
 - Many early approaches for solving games
- Expert Systems
 - hand-coded rules and logic
 - “learn” by humans adding more rules

```
=====
EEEEEEEE L      IIIIII ZZZZZZZ      AAA
E         L      I      Z      A      A
E         L      I      Z      A      A
EEEE     L      I      Z      A      A
E         L      I      Z      A      A
E         L      I      Z      A      A
EEEEEEEE LLLLLLL IIIIII ZZZZZZ      A      A

=====
ELIZA > (PLEASE TYPE IN ALL CAPS) WHAT'S YOUR NAME DEAR ?
AMIT
ELIZA > HI, I'M ELIZA. WHAT DO YOU WANT TO TALK ABOUT ?
AMIT > CAN I TALK ABOUT MY PROBLEM ?
ELIZA > SURE... !
AMIT > DUE TO ON GOING EXAMS IN COLLEGE, I HAVE A LOT OF STRESS.
ELIZA > PLEASE RELAX A LITTLE. SLEEP WELL.
AMIT > THANKS FOR YOUR ADVICE.
ELIZA > NO MENTION.
AMIT > BYE
ELIZA > BYE AND KEEP IN TOUCH...

=====
```

1966, Joseph Weizenbaum

Machine Learning

*"A computer program is said to learn from **experience E** with respect to some class of **tasks T** and **performance measure P** if its performance at tasks in T , as measured by P , improves with experience E ."*

-- Tom M. Mitchell

Experience:

Task:

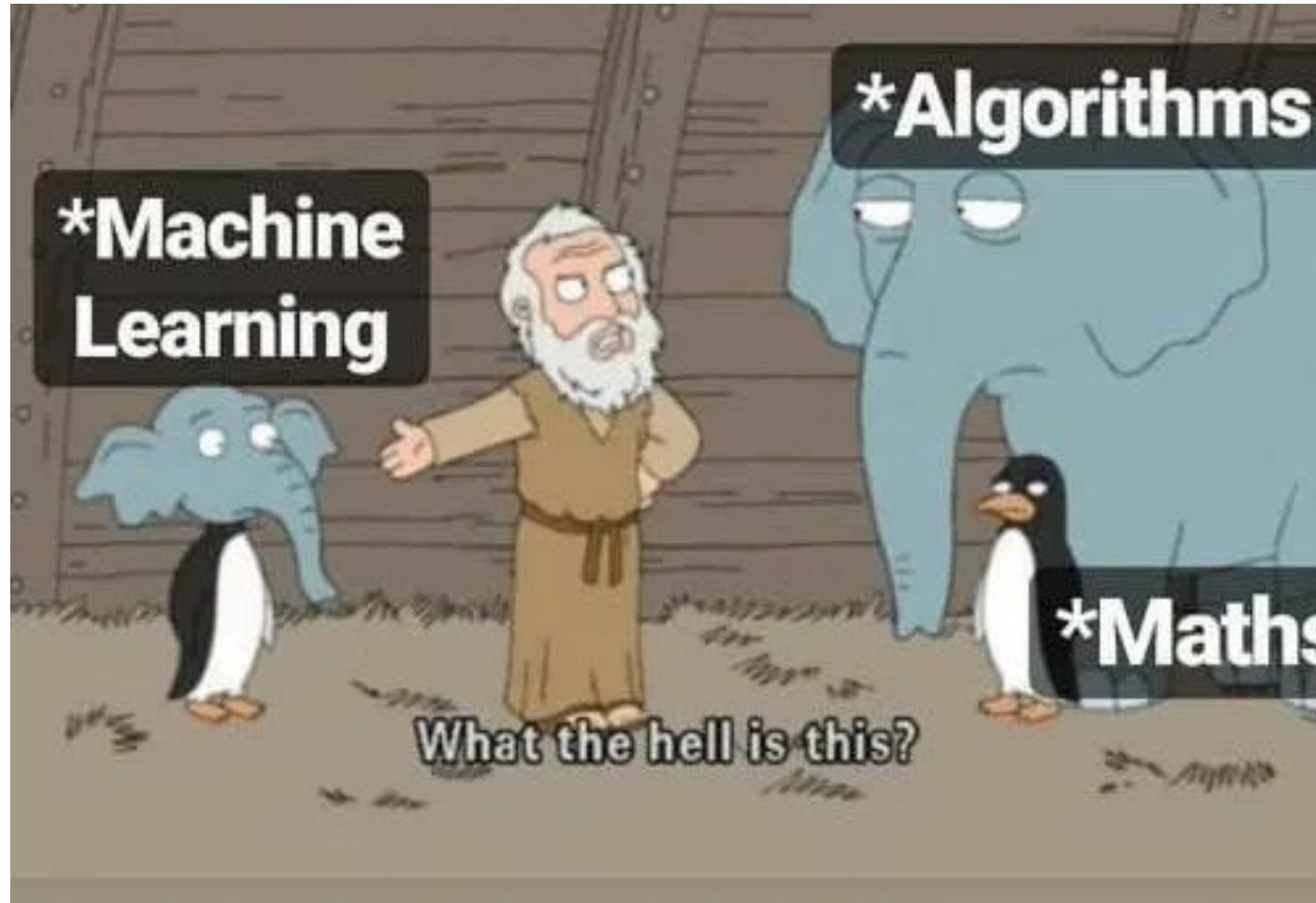
Performance:

Data

Task (classification, recommendation, NL gen, etc.)

Evaluation

What is Machine Learning?



<https://medium.com/nybles/understanding-machine-learning-through-memes-4580b67527bf>

First, let us talk about **Data**

*"A computer program is said to learn from **experience E** with respect to some class of **tasks T** and **performance measure P** if its performance at tasks in T , as measured by P , improves with experience E ."*

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Experience:

Task:

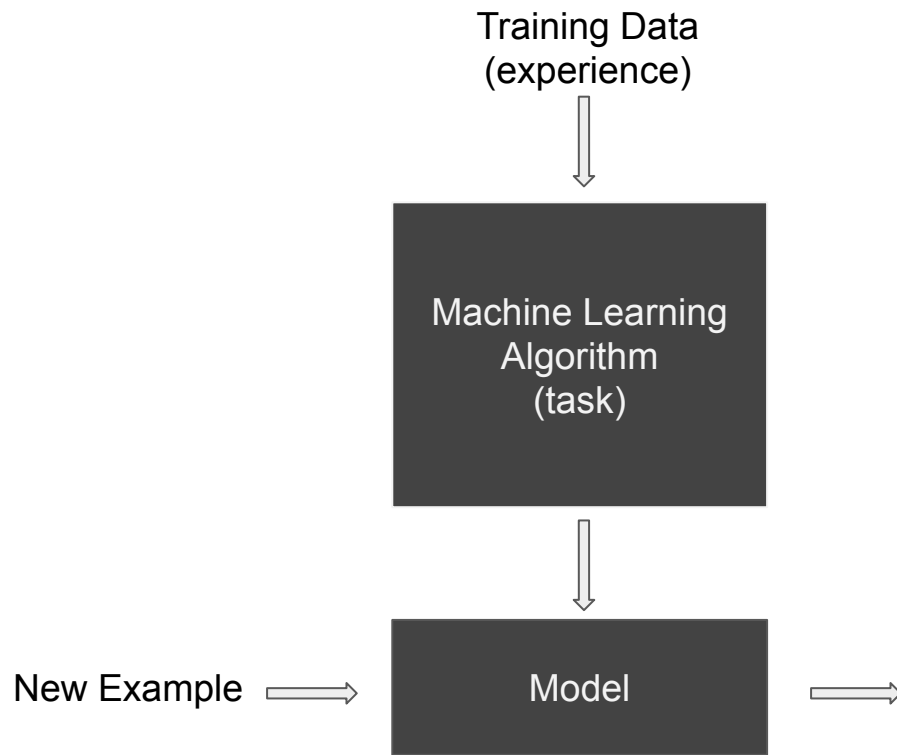
Performance:

Data

Task (classification, recommendation, NL gen, etc.)

Evaluation

Standard Machine Learning Pattern



Standard Machine Learning Pattern



Iris Dataset, UCI Machine Learning Repository

<https://archive.ics.uci.edu/ml/datasets/iris>

5. Number of Instances: 150 (50 in each of three classes)

6. Number of Attributes: 4 numeric, predictive attributes and the class

7. Attribute Information:

1. sepal length in cm

2. sepal width in cm

3. petal length in cm

4. petal width in cm

5. class:

-- Iris Setosa

-- Iris Versicolour

-- Iris Virginica

8. Missing Attribute Values: None

Summary Statistics:

	Min	Max	Mean	SD	Class Correlation
sepal length:	4.3	7.9	5.84	0.83	0.7826
sepal width:	2.0	4.4	3.05	0.43	-0.4194
petal length:	1.0	6.9	3.76	1.76	0.9490 (high!)
petal width:	0.1	2.5	1.20	0.76	0.9565 (high!)

9. Class Distribution: 33.3% for each of 3 classes.

Training Data
(experience)



Machine Learning
Algorithm
(task)



New Example



Model



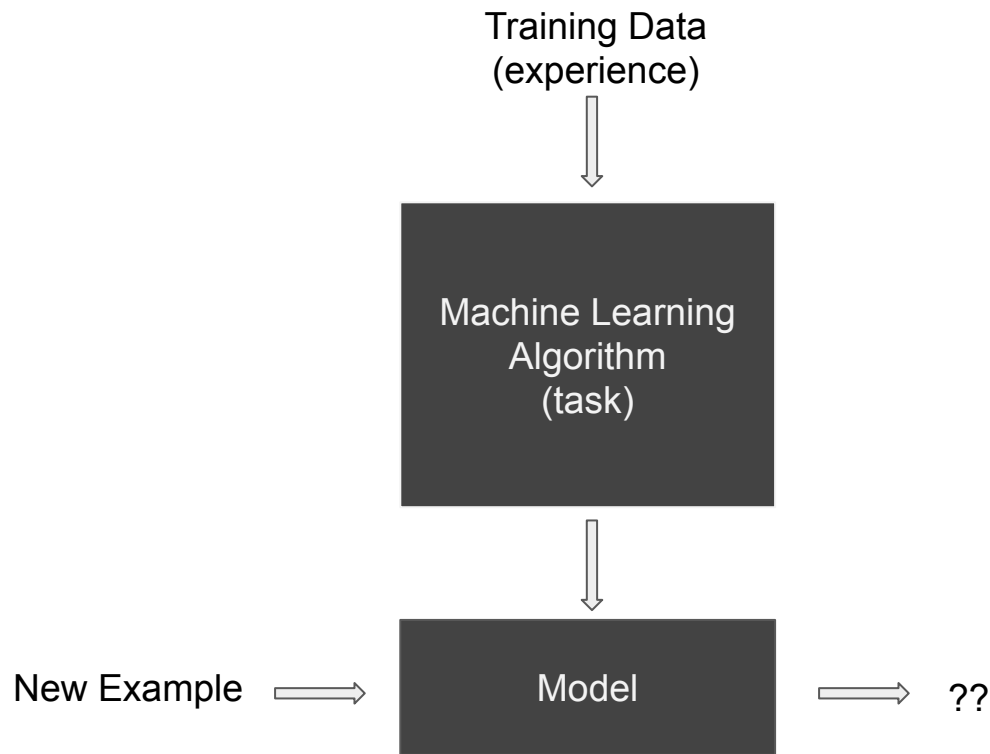
??

Standard Machine Learning Pattern



Iris Dataset, UCI Machine Learning Repository

<https://archive.ics.uci.edu/ml/datasets/iris>



```
5.1,3.5,1.4,0.2,Iris-setosa
4.9,3.0,1.4,0.2,Iris-setosa
4.7,3.2,1.3,0.2,Iris-setosa
4.6,3.1,1.5,0.2,Iris-setosa
5.0,3.6,1.4,0.2,Iris-setosa
5.4,3.9,1.7,0.4,Iris-setosa
4.6,3.4,1.4,0.3,Iris-setosa
5.0,3.4,1.5,0.2,Iris-setosa
4.4,2.9,1.4,0.2,Iris-setosa
4.9,3.1,1.5,0.1,Iris-setosa
5.4,3.7,1.5,0.2,Iris-setosa
4.8,3.4,1.6,0.2,Iris-setosa
4.8,3.0,1.4,0.1,Iris-setosa
4.3,3.0,1.1,0.1,Iris-setosa
5.8,4.0,1.2,0.2,Iris-setosa
5.7,4.4,1.5,0.4,Iris-setosa
```

Datasets

Documentation

New Dataset

Public

Sort by Hotness

10,252 Datasets

Sizes

File types

Licenses

Tags

Search datasets

163

**Telco Customer Churn**

Focused customer retention programs

BlastChar updated 7 months ago (Version 1)

telecommu...
churn analy...CSV
171.6 KB
Other</> 80
3
23k

44

**Weather Conditions in World War Two**

Daily Weather Summaries from 1940-1945

Shane Smith updated 10 months ago (Version 1)

weather
history
warCSV
1.6 MB
Other</> 8
0
11k

72

**Education Statistics**

From World Bank Open Data

World Bank updated a day ago (Version 37)

world
countries
educationCSV
75.1 MB
Other</> 48
0
25k

148

**Avocado Prices**

Historical data on avocado prices and sales volume in multiple US markets

Justin Kiggins updated 3 months ago (Version 1)

food and dr...

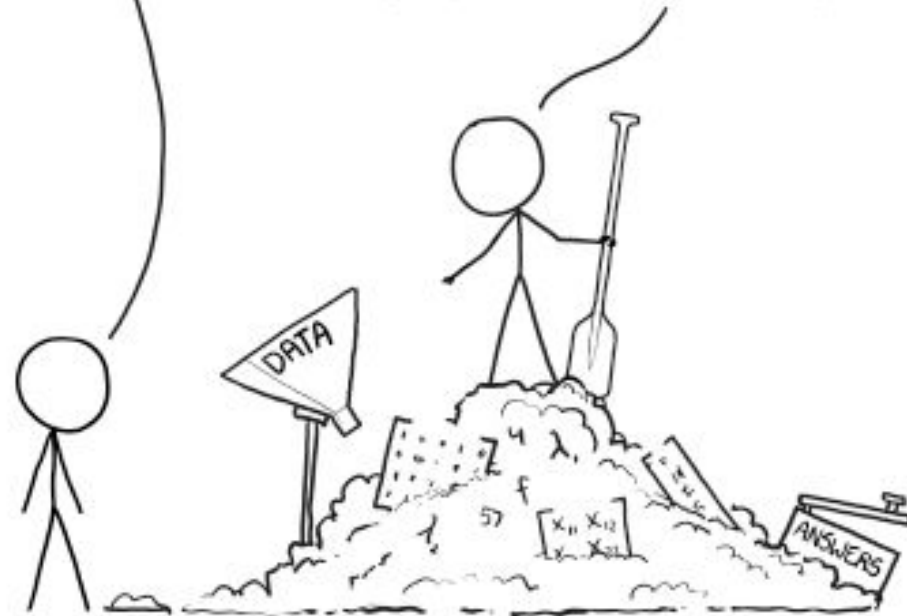
CSV
628.7 KB
ODbL</> 39
4
26k

THIS IS YOUR MACHINE LEARNING SYSTEM?

YUP! YOU POUR THE DATA INTO THIS BIG
PILE OF LINEAR ALGEBRA, THEN COLLECT
THE ANSWERS ON THE OTHER SIDE.

WHAT IF THE ANSWERS ARE WRONG?

JUST STIR THE PILE UNTIL
THEY START LOOKING RIGHT.





What tasks would you send an island full of workers with laptops? Imagine they're far away and don't understand instructions, but eager to learn by example.

Machine Learning

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-- Tom M. Mitchell

Experience:

Task:

Performance:

Data

Task (classification, recommendation, NL gen, etc.)

Evaluation

Next, Task

*"A computer program is said to learn from **experience E** with respect to some class of **tasks T** and **performance measure P** if its performance at tasks in T, as measured by P, improves with experience E."*

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Evaluation

Terminology: Machine learning tasks

- Classification
 - Is this a dog or a cat?
- Regression
 - How warm will it be tomorrow?
- Clustering
 - Here are a few articles. Organize them into topics
- Representation
 - What is the best way to represent words? (e.g. so representation of “autumn” and “fall” are similar)
 - What is the color of happiness?
- “Dear algorithm: figure it out, and if you’re right, you get a reward”
 - Drive a car
 - Recommend movies to people

Machine learning tasks

- Classification

- Is this a dog or a cat?

- Regression

- How warm will it be tomorrow?

Supervised learning

- Clustering

- Here are a few articles. Organize them into topics

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Unsupervised learning

- “Dear algorithm: figure it out, and if you’re right, you get a reward”

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Reinforcement learning

Difference between supervised and reinforcement learning

- In reinforcement learning, the data you collect depends on your previous actions
 - If you show an ad, you get to see if the user clicks on it
 - But you don't get to see if the user may also have clicked on a different ad if you had shown that instead
- One way to measure the performance of your reinforcement learning algorithm is to use “regret”:
 - How much do you lose compared to the best solution?
 - e.g. if you knew exactly what the user wanted, what ads would you show them?

Now, let's look at our “friend”, ChatGPT

We introduce the training process of a language model using reinforcement learning in three steps:

Step 1

Collect demonstration data and train a supervised policy.

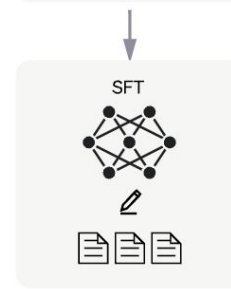
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



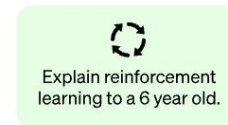
This data is used to fine-tune GPT-3.5 with supervised learning.



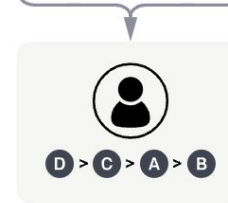
Step 2

Collect comparison data and train a reward model.

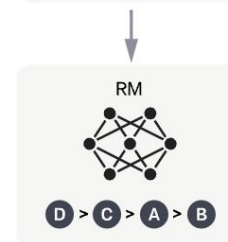
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



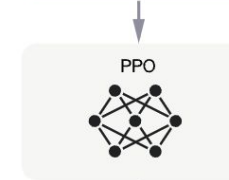
Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

A new prompt is sampled from the dataset.



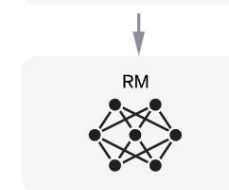
The PPO model is initialized from the supervised policy.



The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



Reward Function

A *rule* defined by the environment that tells the agent how good or bad a state or action is.

Role: Provides feedback in the form of scalar rewards (numbers).

Who defines it: The *designer* of the problem, not the agent.

Example: In a robot navigation task: -1 for hitting a wall, +10 for reaching the goal, -0.1 for each step taken (to encourage efficiency).

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Policy

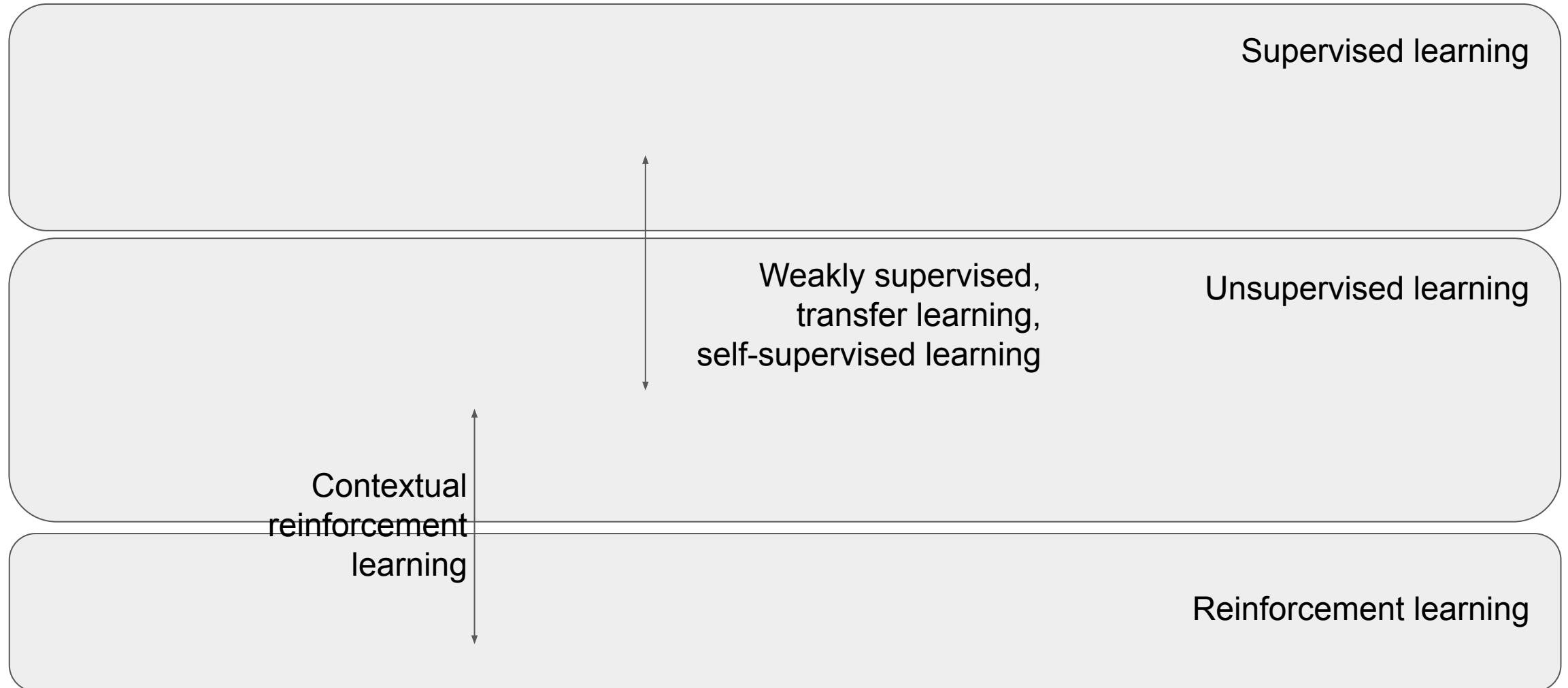
strategy or mapping from states (and sometimes history) to actions.

Role: Describes how the agent behaves—what action it takes given the current situation.

Who learns it: The agent learns or improves the policy through interaction with the environment and the rewards it gets.

Example: In robot navigation: “If near the goal, move forward; if facing an obstacle, turn left.”

Machine learning tasks



Could you share more examples of machine learning tasks?

- Classification

Supervised learning

- Is this a dog or a cat?

- Regression

- How warm will it be tomorrow?

- Clustering

Unsupervised learning

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Reinforcement learning

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And Evaluation

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Experience:

Task:

Performance:

Data

Task (classification, recommendation, NL gen, etc.)

Evaluation

Terminology: Evaluation

- What metrics do we use to discuss performance?
- How do we set up a meaningful evaluation?

Evaluation Metrics

- We're generally interested in the following:
 - How often is the prediction wrong?
 - How is the prediction wrong?
 - What is the cost of wrong predictions?
 - How can we minimize cost? (or regret?)

Accuracy

- correct predictions/total predictions

Accuracy in Speech Recognition

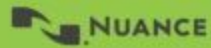
Nuance unveils AI-powered Dragon® Medical Practice Edition 4

Next generation speech engine achieves **99% out-of-the-box accuracy** without voice profile training, boosting clinician efficiency while eliminating transcription costs.

Create narrative documentation directly in EHR fields faster and more accurately than ever before.



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a “98% accurate” spam filter

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Democrats

● This article is more than 2 years old

Top Democrat's emails hacked by Russia after aide made typo, investigation finds

In the run-up to the US election, aide to John Podesta spotted phishing email but flagged it as 'legitimate' instead of 'illegitimate'



▲ The revelations give further credence to the CIA finding that the Kremlin intervened to help Donald Trump defeat Hillary Clinton. Photograph: Jewel Samad/AFP/Getty Images

From: Charles Delavan <cdelavan@hillaryclinton.com>

Date: March 19, 2016 at 9:54:05 AM EDT

To: Sara Latham <slatham@hillaryclinton.com>, Shane Hable <shable@hillaryclinton.com>

Subject: Re: Someone has your password

Sara,

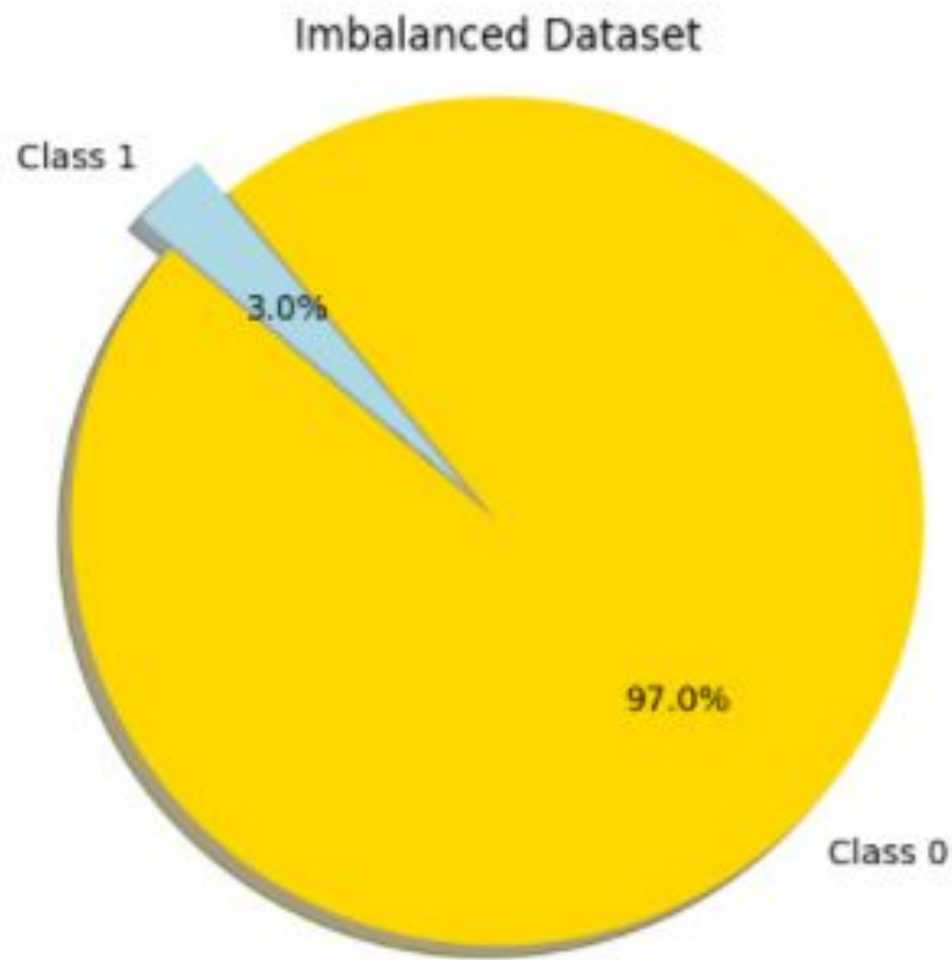
This is a legitimate email. John needs to change his password immediately, and ensure that two-factor authentication is turned on his account.

He can go to this link: <https://myaccount.google.com/security> to do both. It is absolutely imperative that this is done ASAP.

Why **Accuracy** isn't enough.



Why **Accuracy** isn't enough



Metrics: What do we measure?

What can happen when your model classifies something (e.g. do you have the COVID virus?)

		<i>Prediction</i>	
		Infected (positive)	Healthy (negative)
<i>Ground Truth</i>	Infected	true positives (TP)	false negative (FN)
	Healthy	false positives (FP)	true negative (TN)

Exercise: More examples

Build a classifier to predict the likelihood that a loan applicant will pay back the loan (or not): Assignment 1

<i>Ground Truth</i>		Will pay back (positive)	Will not pay back (negative)
	Actually pay back	true positives (TP)	false negative (FN)
	Actually not pay back	false positives (FP)	true negative (TN)

All Errors Are Not Equal

Depending on your task, different errors have different costs.

Cost of “false negative” in COVID tests?

Cost of “false positive” in COVID test?

In law enforcement?

In Cancer screening?

In detecting the “Alexa” command?

In detecting person in the road?

Metrics: Terminology

Precision

$$\frac{tp}{tp + fp}$$

- out of all the examples that *predicted* positive, how many are actually positive?

Recall/Sensitivity == True Positive Rate

$$\frac{tp}{tp + fn}$$

- out of all *actual* positive cases, how many are predicted as positive?
- **ability of the test to identify those who have a disease**

Specificity == True Negative Rate

$$\frac{tn}{tn + fp}$$

- out of all *actual* negative cases, how many are predicted as negative?
- **ability of the test to identify those without the disease**

Johnson

In the world of voice-recognition, not all accents are equal

But you can train your gadgets to understand what you're saying



<https://www.economist.com/books-and-arts/2018/02/15/in-the-world-of-voice-recognition-not-all-accent-are-equal>

Amazon's Face Recognition Falsely Matched 28 Members of Congress With Mugshots



By [Jacob Snow](#), Technology & Civil Liberties Attorney, ACLU of Northern California

JULY 26, 2018 | 8:00 AM

TAGS: [Face Recognition Technology](#), [Surveillance Technologies](#), [Privacy & Technology](#)



Amazon's face surveillance technology is the target of growing opposition nationwide, and today, there are 28 more causes for concern. In a test the ACLU recently conducted of the facial recognition tool, called "Rekognition," the software incorrectly matched 28 members of Congress, identifying them as other people who have been arrested for a crime.

The members of Congress who were falsely matched with the mugshot database we used in the test include Republicans and Democrats, men and

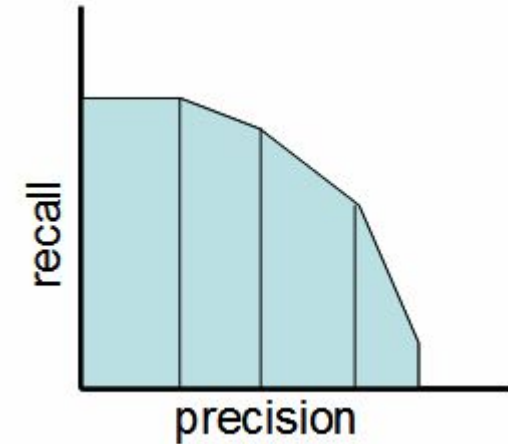


The false matches were disproportionately of people of color, including six members of the Congressional Black Caucus, among them civil rights legend Rep. John Lewis (D-Ga.). These results demonstrate why Congress should join the ACLU in calling for a moratorium on law enforcement use of face surveillance.

Question?

Can you **provide an example where different types of mistakes may have very different costs?**

Metrics: Precision and Recall



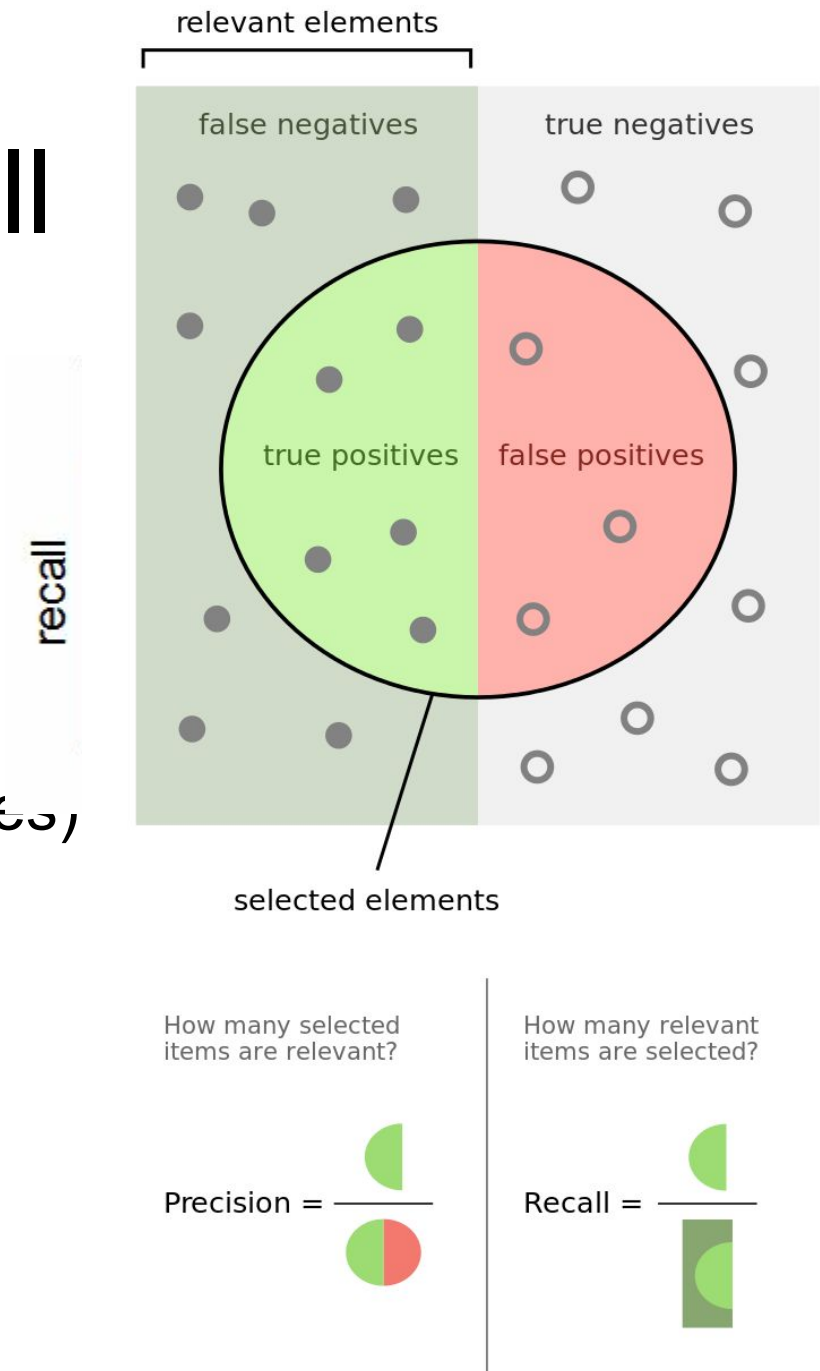
Precision = true positives / (true positives + false positives)

Recall = true positives / (true positives + false negatives)

Metrics: Precision and Recall

Precision = true positives / (true positives + false positives),

Recall = true positives / (true positives + false negatives)



Metrics: F1 score

Often used in information retrieval

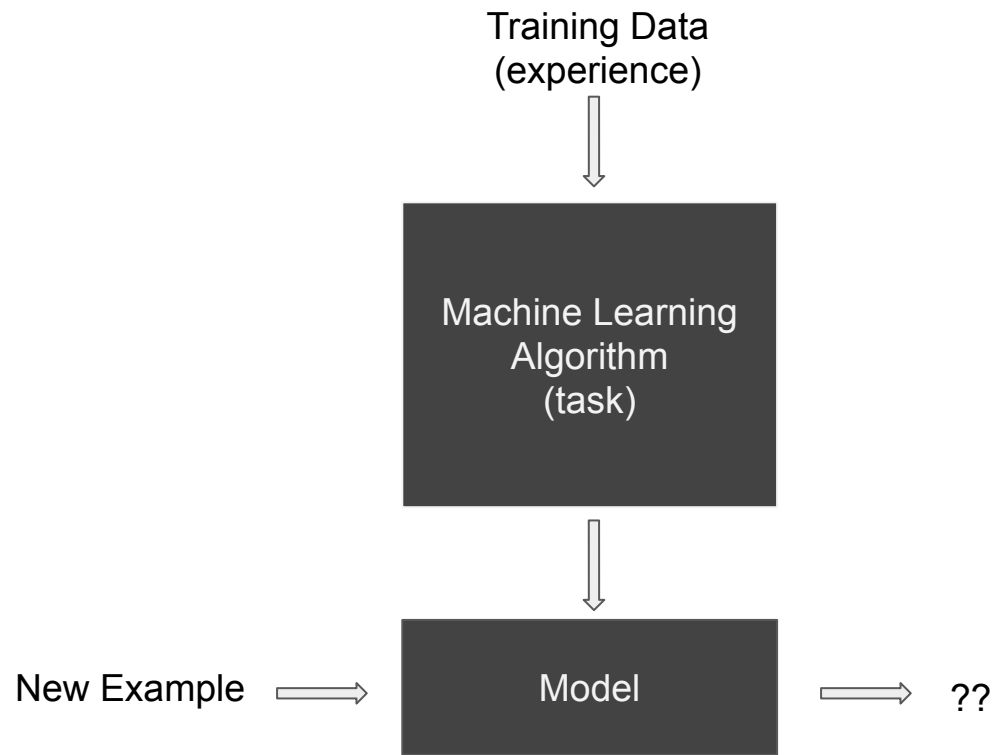
- The harmonic mean of precision and recall.

$$F_1 = \left(\frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} \right) = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

Silent errors

- Not all errors are visible to the computer
- How do you measure when your device can't hear "OK, Google"

The “standard textbook model”

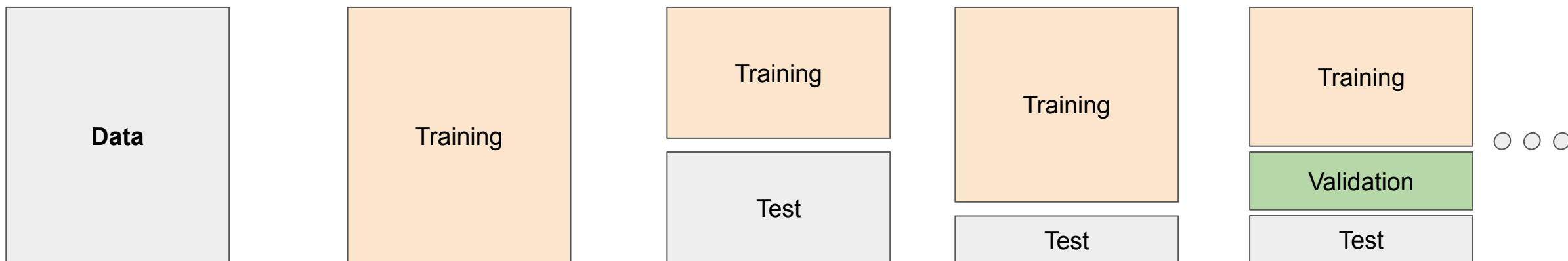


Setting Up Meaningful Evaluations

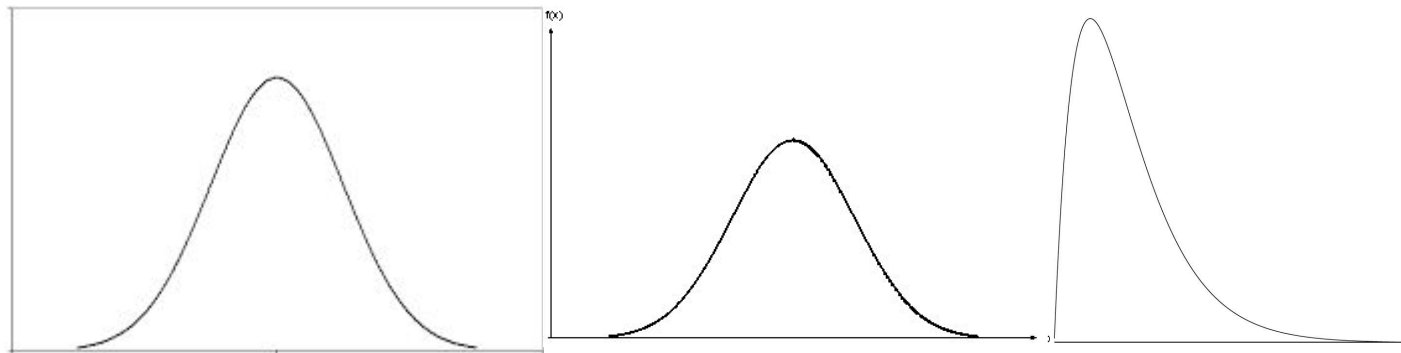
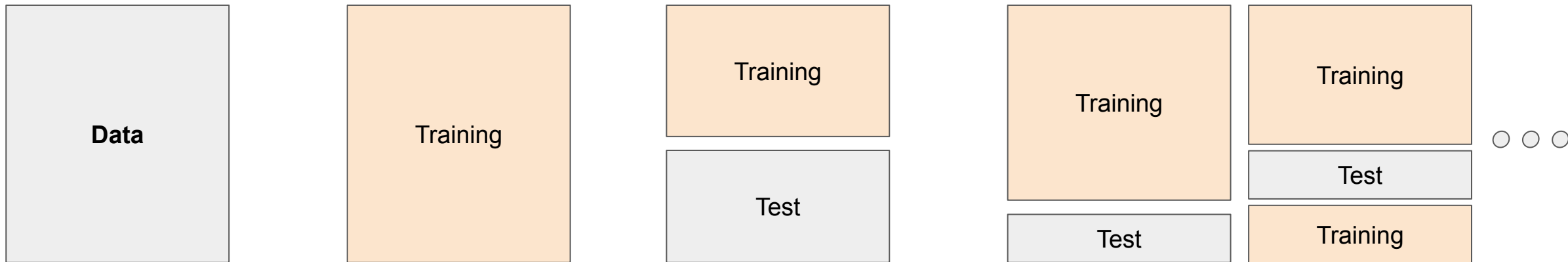
- *Avoiding Overfitting*

Easy for models to learn their own training data

Train, Test, and Validate



Train, Test, and Validate





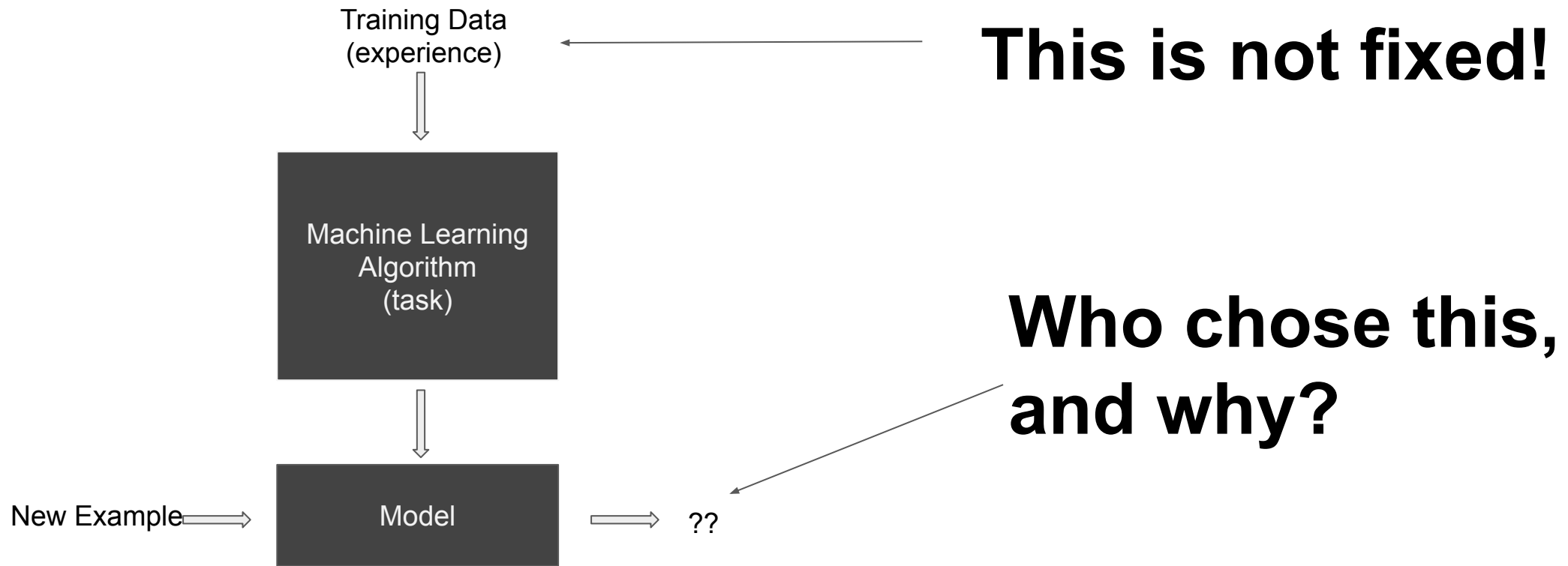
What kind of error is this?

Jumping out a level...

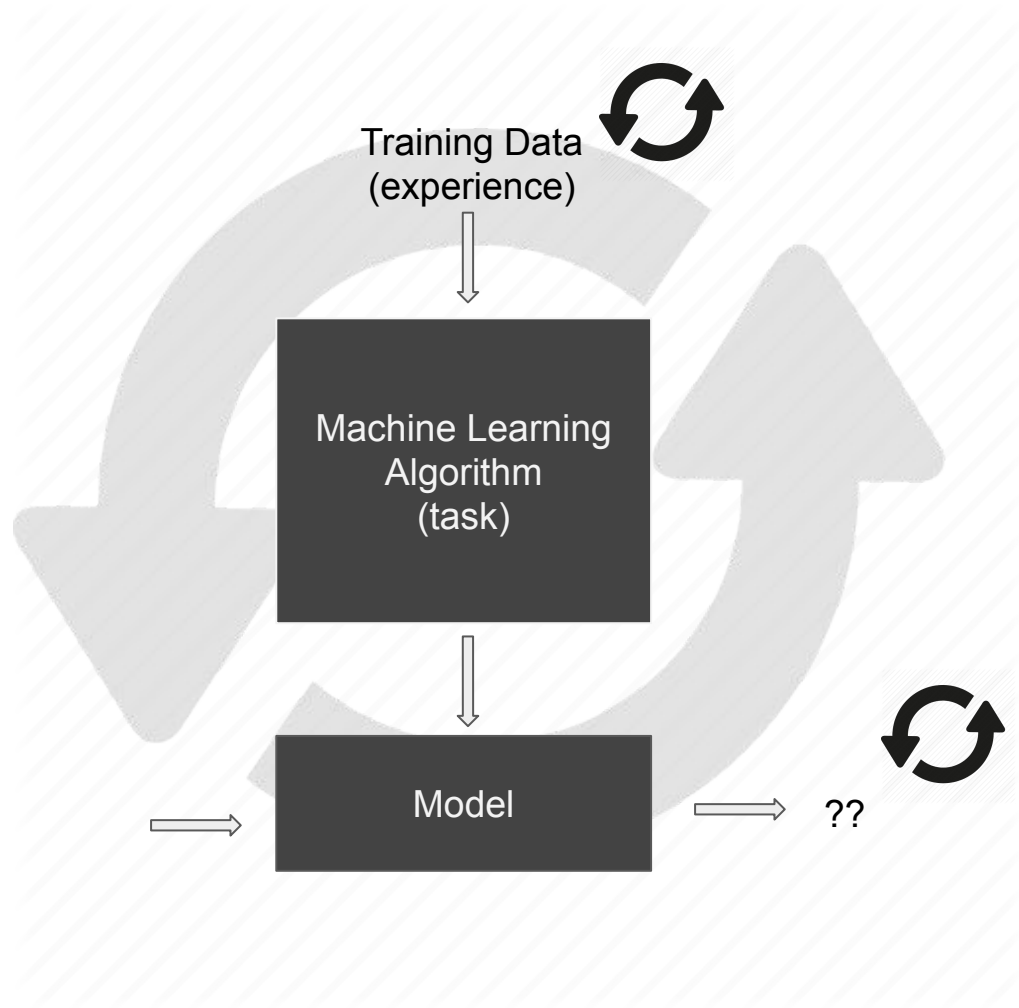
The flip side of overfitting... failure to generalize, failure to adapt:

- What happens when your model goes out into the world?

The Unquestioned Textbook Assumption

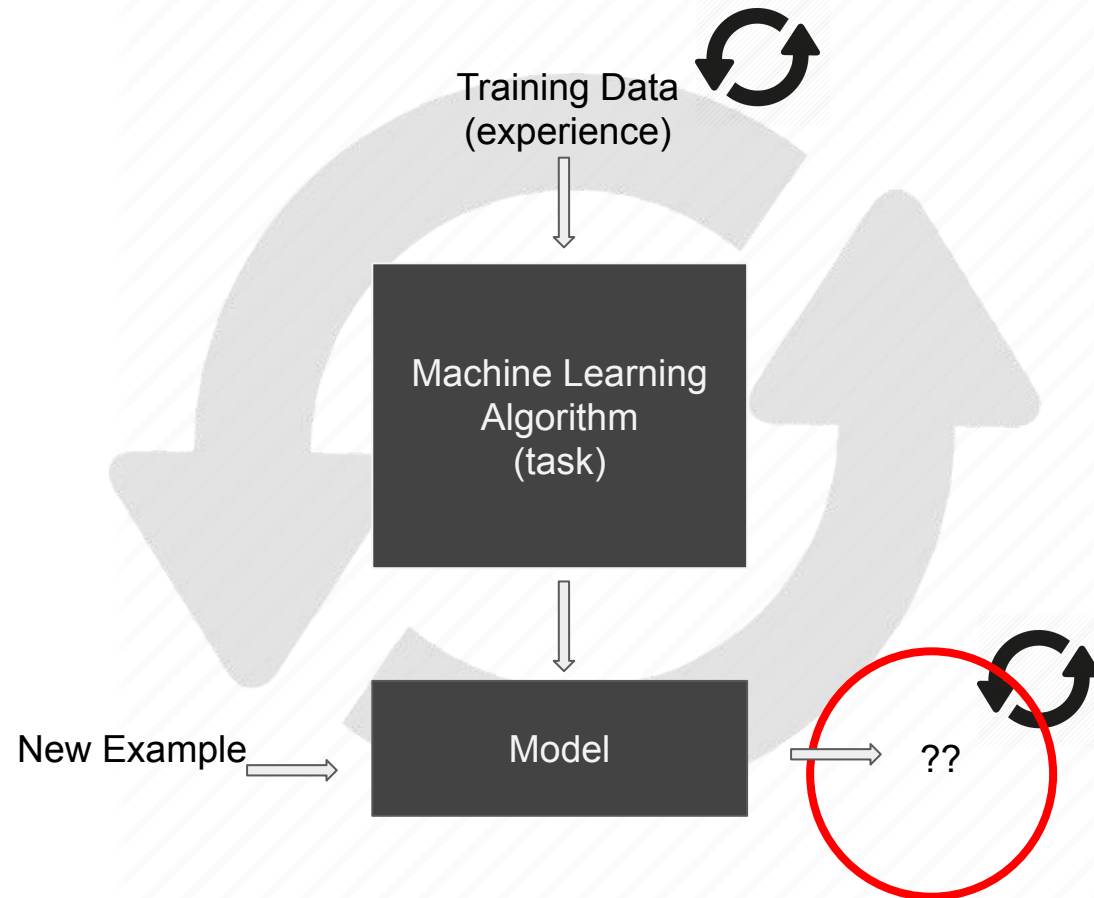


Iterative not Waterfall



- everything can and should be iterated on, including the problem itself ... what are you trying to solve?

Start at the End ...



Connection to Direct Manipulation

- Continuous Representation of Object of Interest (feedback)
- Rapid, incremental, and reversible interactions
- Physical Actions

Connection to Direct Manipulation (Agents)

- Continuous Representation of Object of Interest (feedback)
 - * agents: continuous representation not possible
- Rapid, incremental, and reversible interactions
 - * agents: doesn't know how it got there, can't go back
- Physical Actions
 - * agents: tendency to force the unexpected

Summary

- Building Human-AI Systems is Iterative
 - resist the “Pipeline” metaphor
- Terminology and Way of Thinking About Performance/Errors
 - these apply regardless of what you stick in the black box
- A major issue in HCI for AI is dealing w/ uncertainty and errors
 - We will talk about this in this course

Any Questions?

Reminder

- Assignment 1 is released
- Next readings reflection due next Wed noon